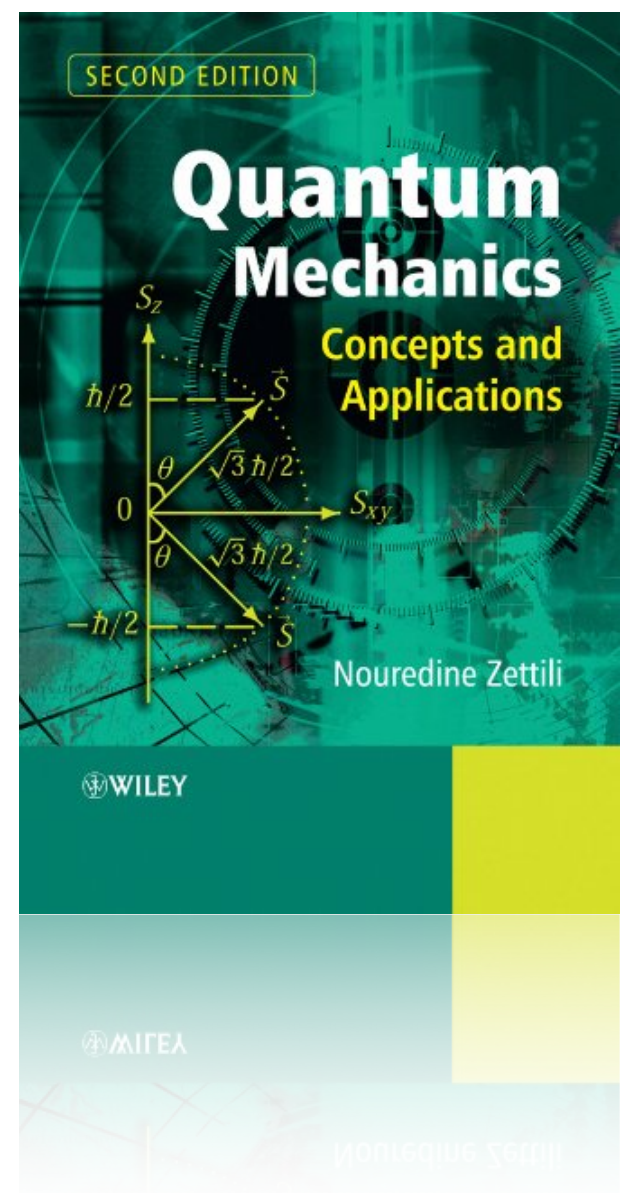


Origin of Quantum Physics

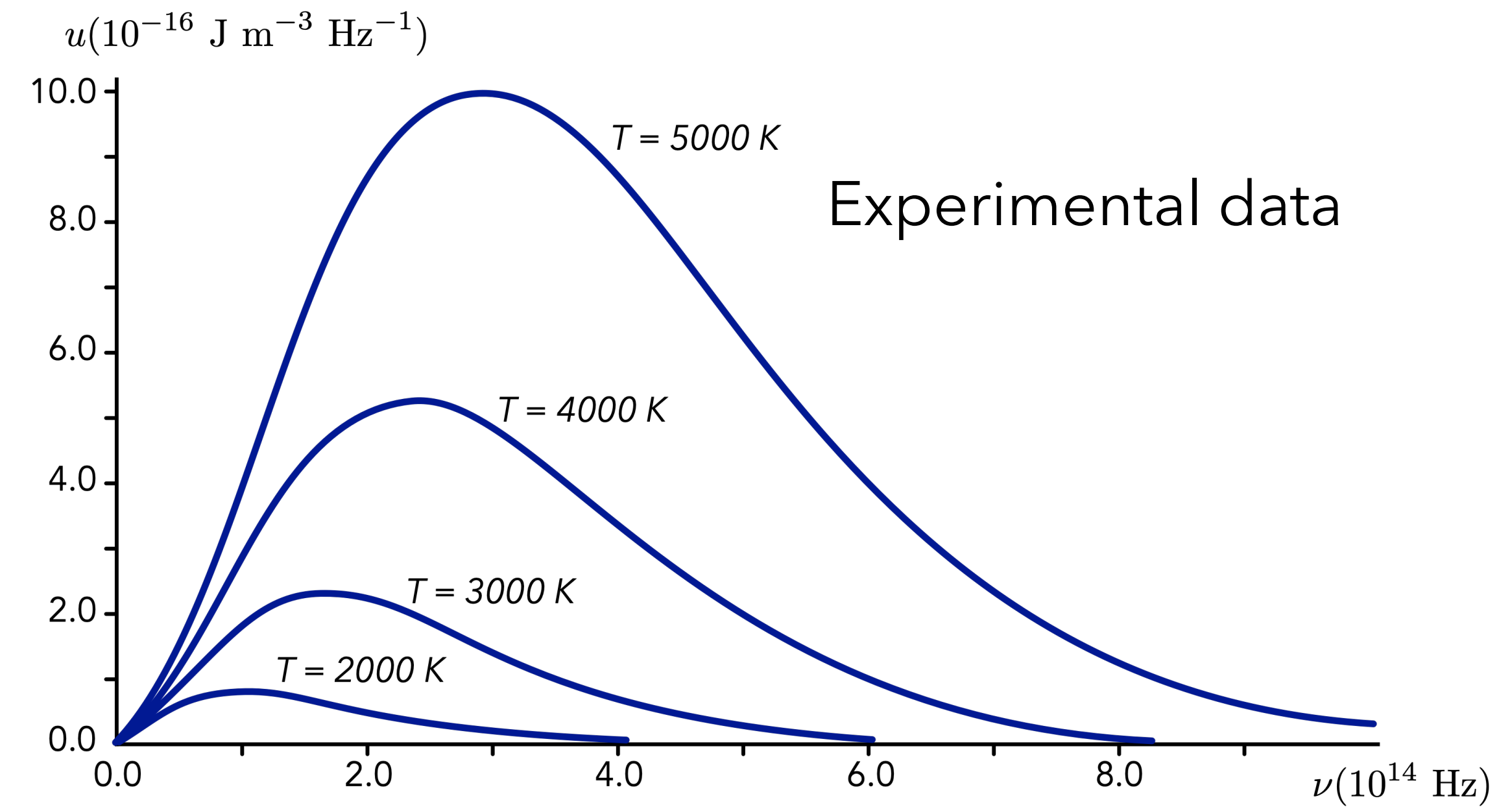
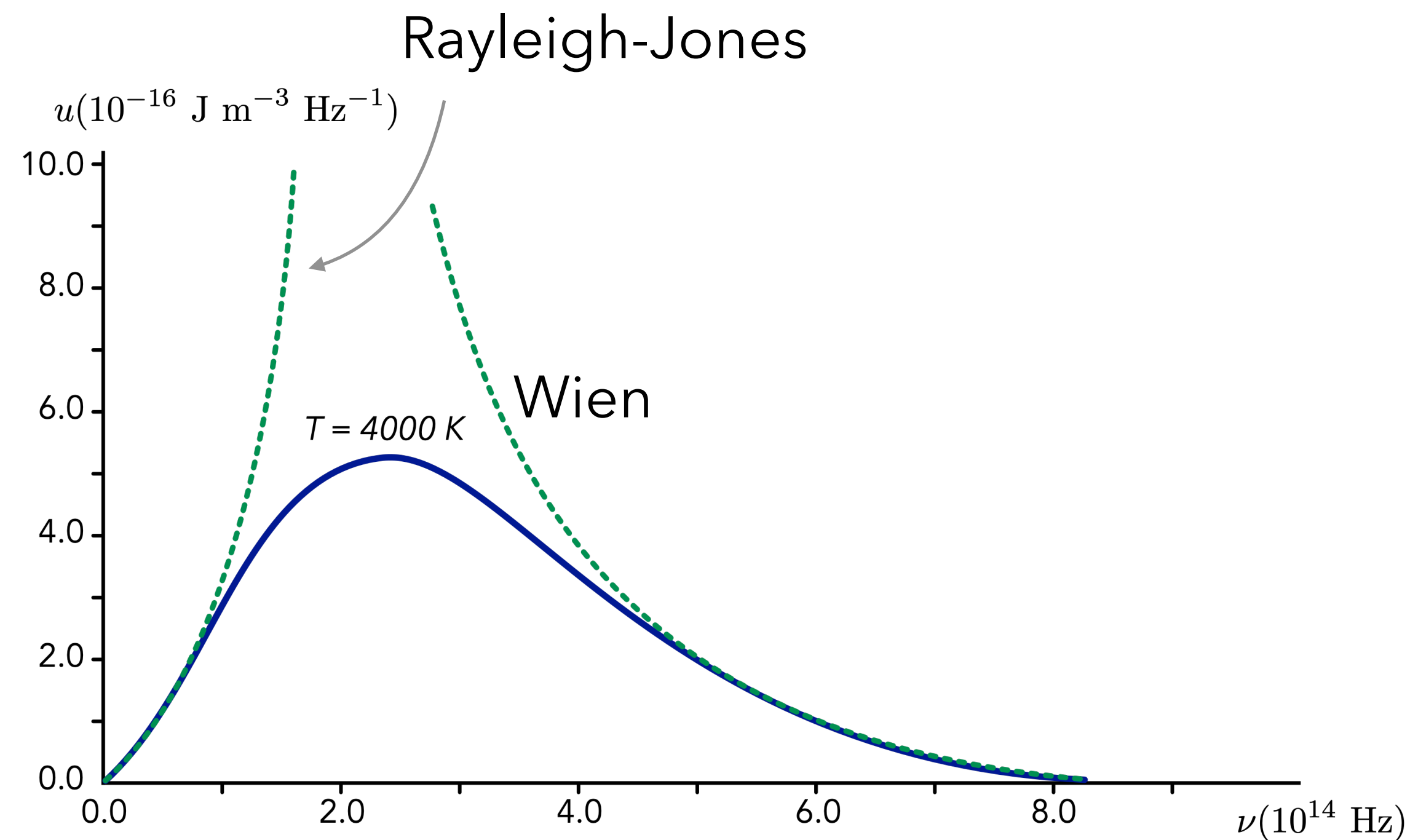
- Historical overview
- Localization - wave packets
- Time evolution of wave packets



Black-body radiation



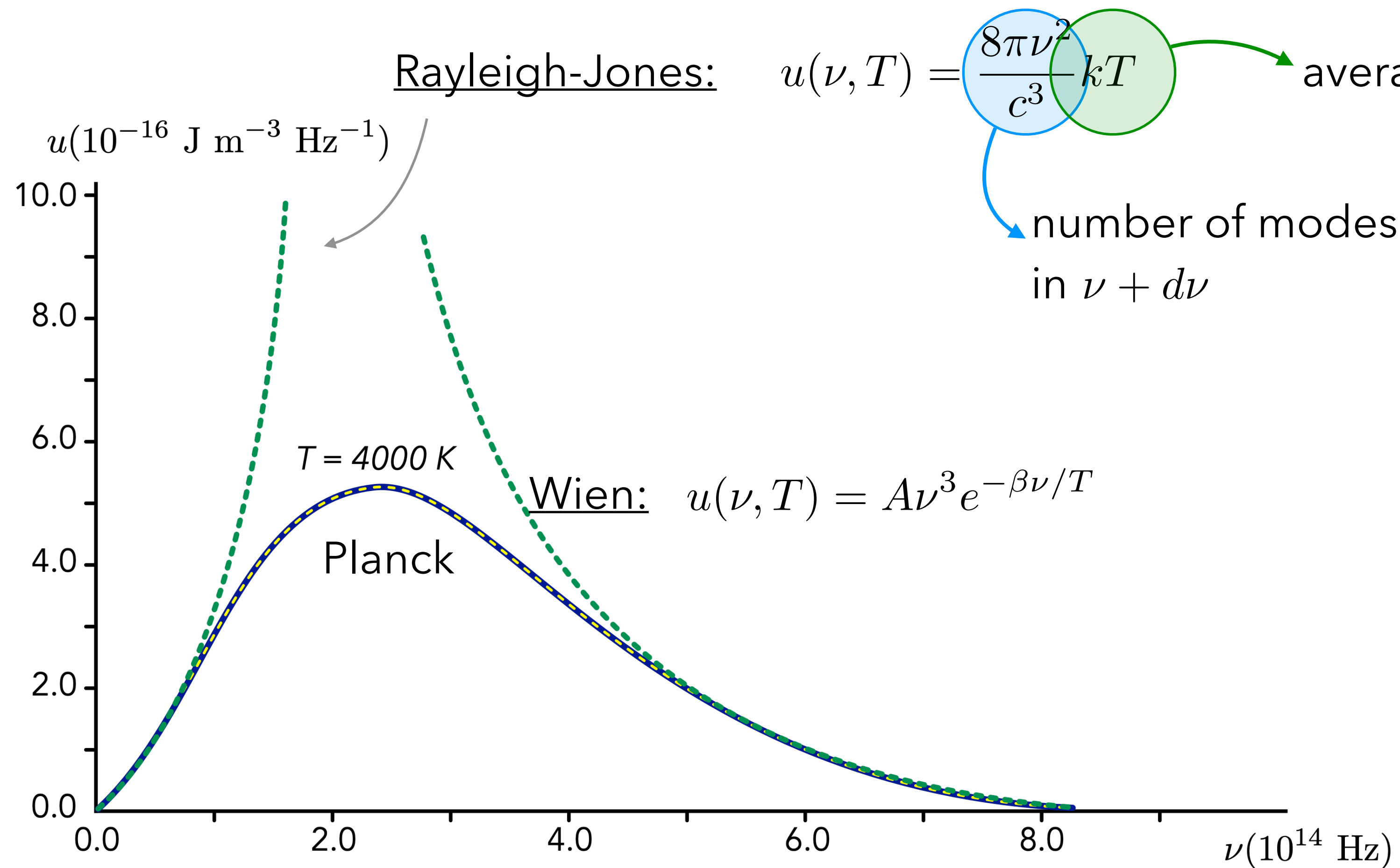
Color depends on the temperature
red $\rightarrow$ yellow $\rightarrow$ white



Wien: $u(\nu, T) = A\nu^3 e^{-\beta\nu/T}$

Rayleigh-Jones: $u(\nu, T) = \frac{8\pi\nu^2}{c^3} kT$

Black-body radiation



$$\langle E \rangle = \frac{\int_0^\infty E e^{-E/kT} dE}{\int_0^\infty e^{-E/kT} dE} = kT$$

Planck (1900): $E = nh\nu, \quad n = 1, 2, 3, \dots$

$$\langle E \rangle = \frac{\sum_{n=0}^\infty nh\nu e^{-nh\nu/kT}}{\sum_{n=0}^\infty e^{-nh\nu/kT}} = \frac{h\nu}{e^{h\nu/kT} - 1}$$

$$u(\nu, T) = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{h\nu/kT} - 1}$$

exact fitting with $h = 6.62607015 \cdot 10^{-34} \text{ m}^2 \text{ kg/s}$

Photoelectric effect

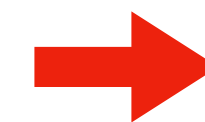
➔ $\nu_{in} < \nu_0$

1. No electrons are emitted

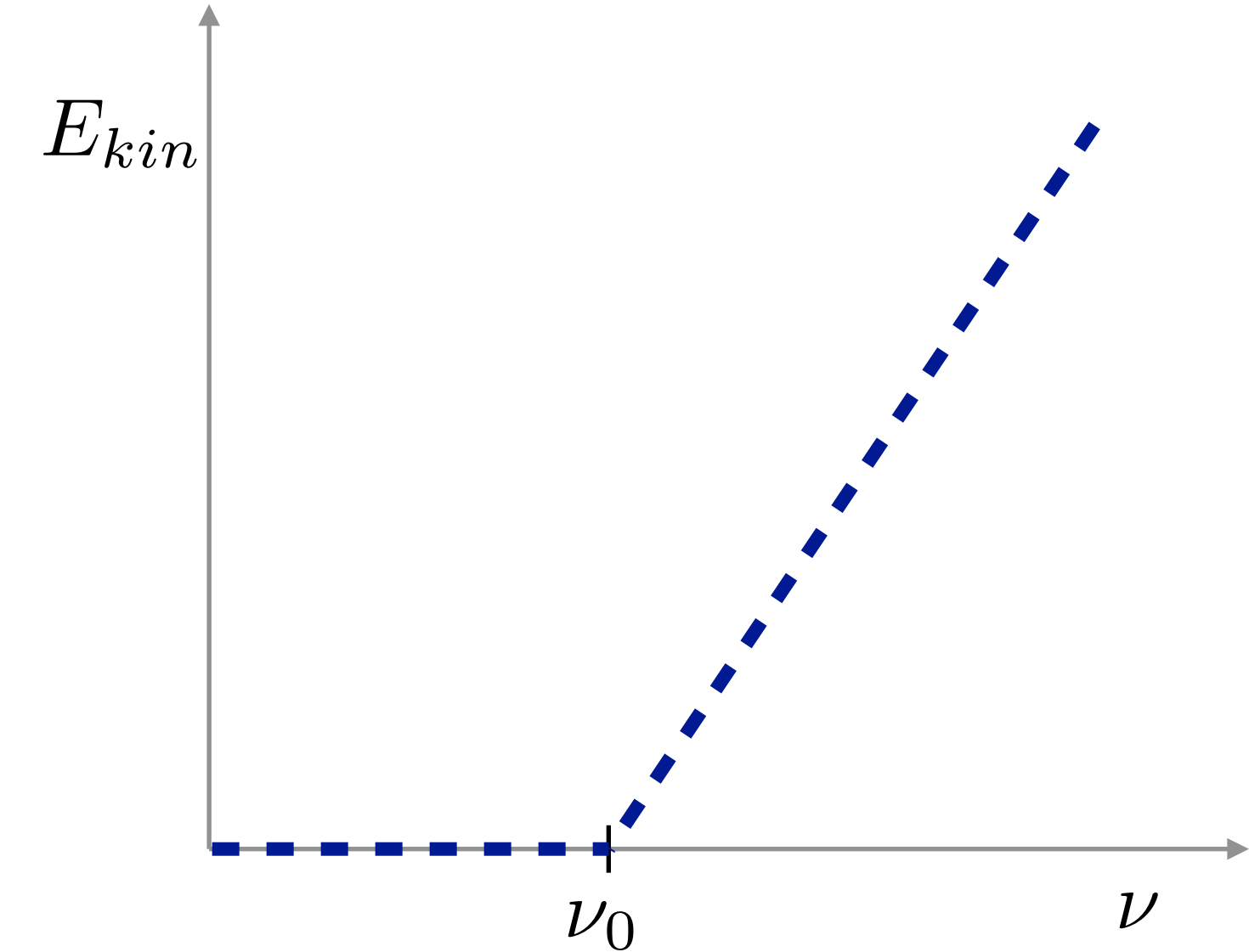
➔ $\nu_{in} > \nu_0$

2. Instantaneous emission, independent of the intensity of the incoming radiation
3. Number of electrons is proportional to intensity
4. Kinetic energy depends on the frequency, not on the intensity

Classical physics: Any amount of energy can be exchanged with matter.



1. Radiation of any frequency with sufficient intensity can provide the necessary energy to liberate electrons
2. Radiation of weak intensity slowly builds up the energy for an electron to escape



Photoelectric effect

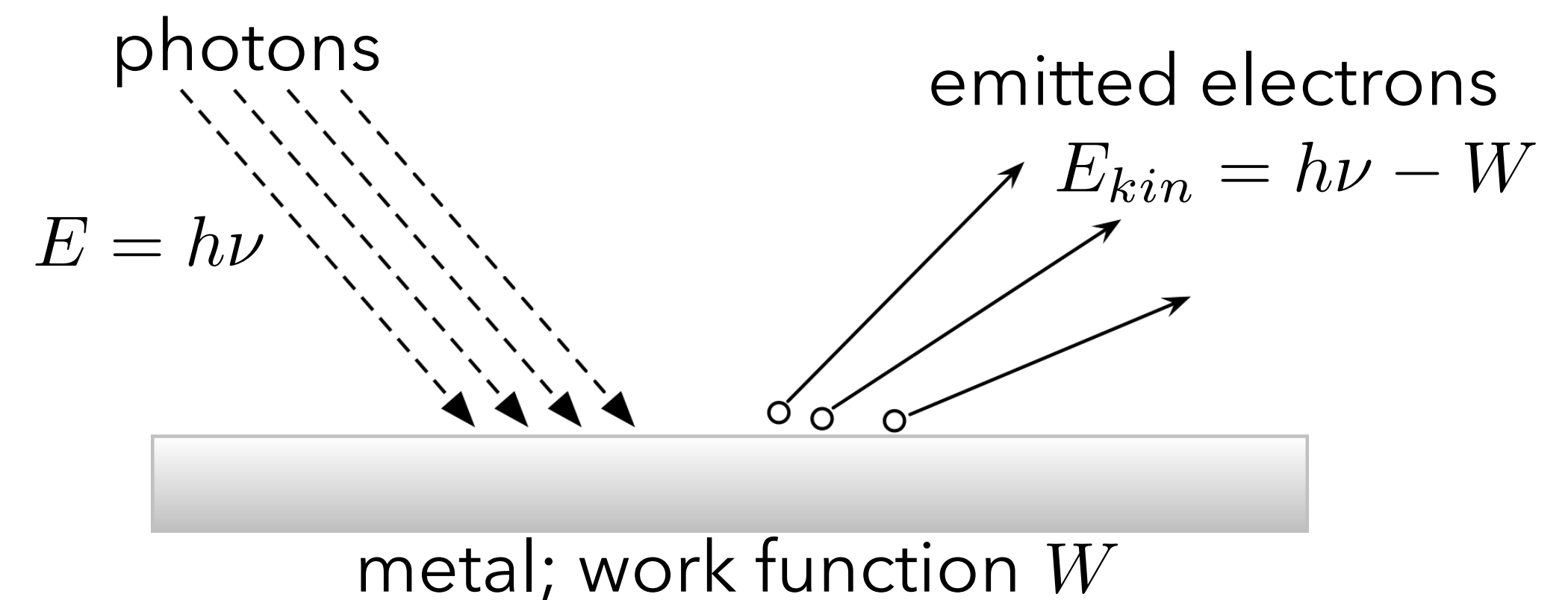
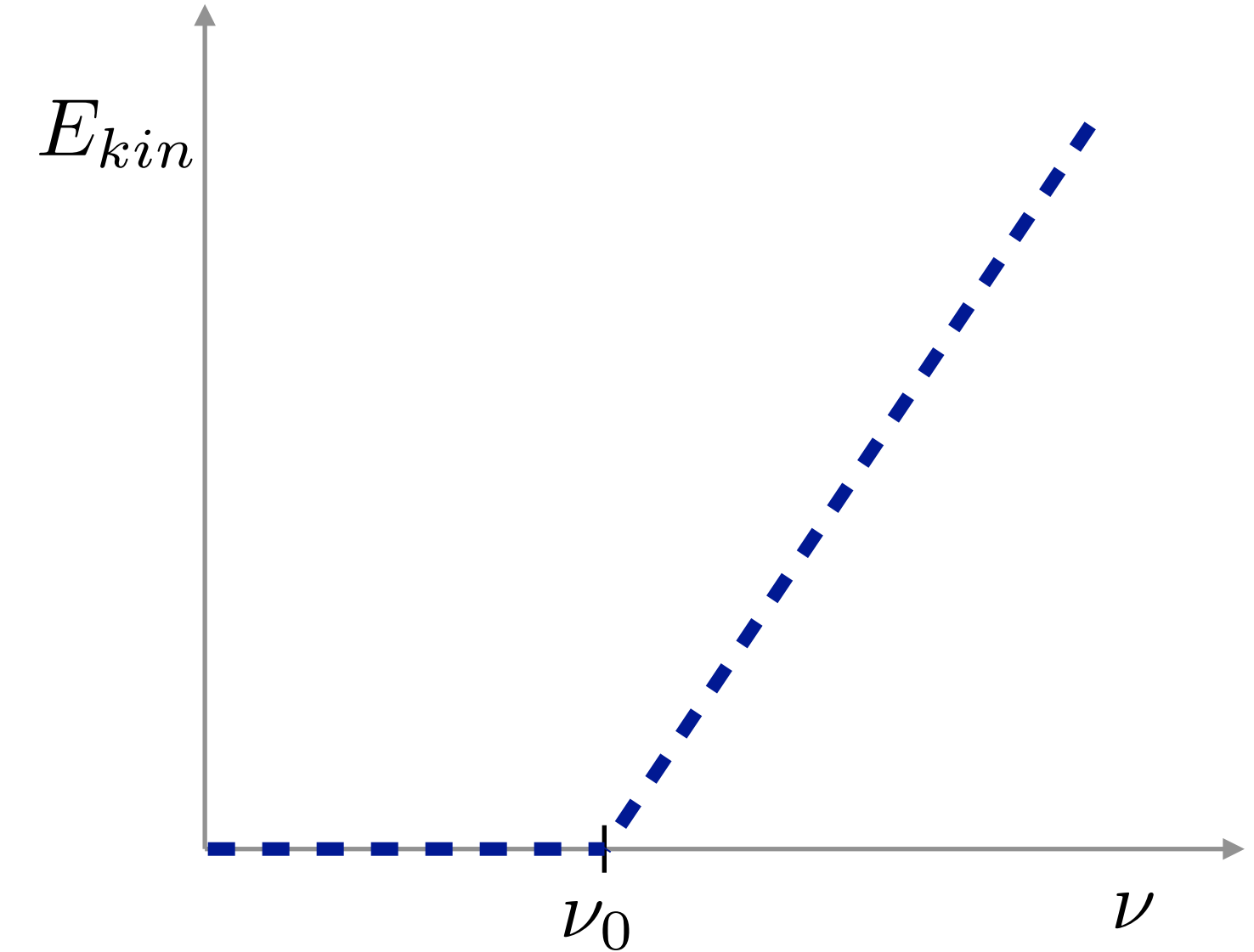
➔ $\nu_{in} < \nu_0$

1. No electrons are emitted

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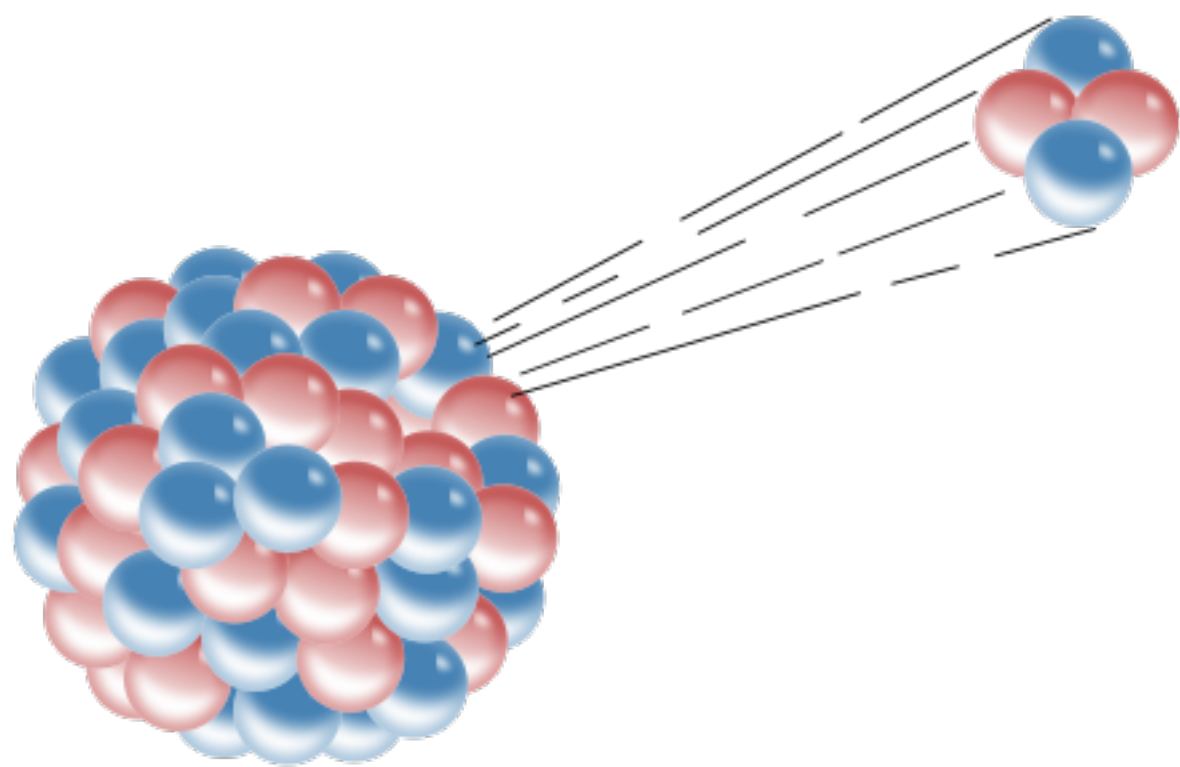
Quantum physics: Light particles, each carrying an energy of $h\nu$ (**Einstein, 1905**)



Wave character of matter

photon: $p = \frac{h\nu}{c} = \frac{h}{\lambda}$

de Broglie (1924): All particles with momentum p can behave as waves with wavelength $\lambda = \frac{h}{p}$



α -particle with $E_{kin} = 5 \text{ MeV}$

$$m = 6.64 \times 10^{-27} \text{ kg}$$

$$\text{size: } 1.68 \times 10^{-15} \text{ m}$$

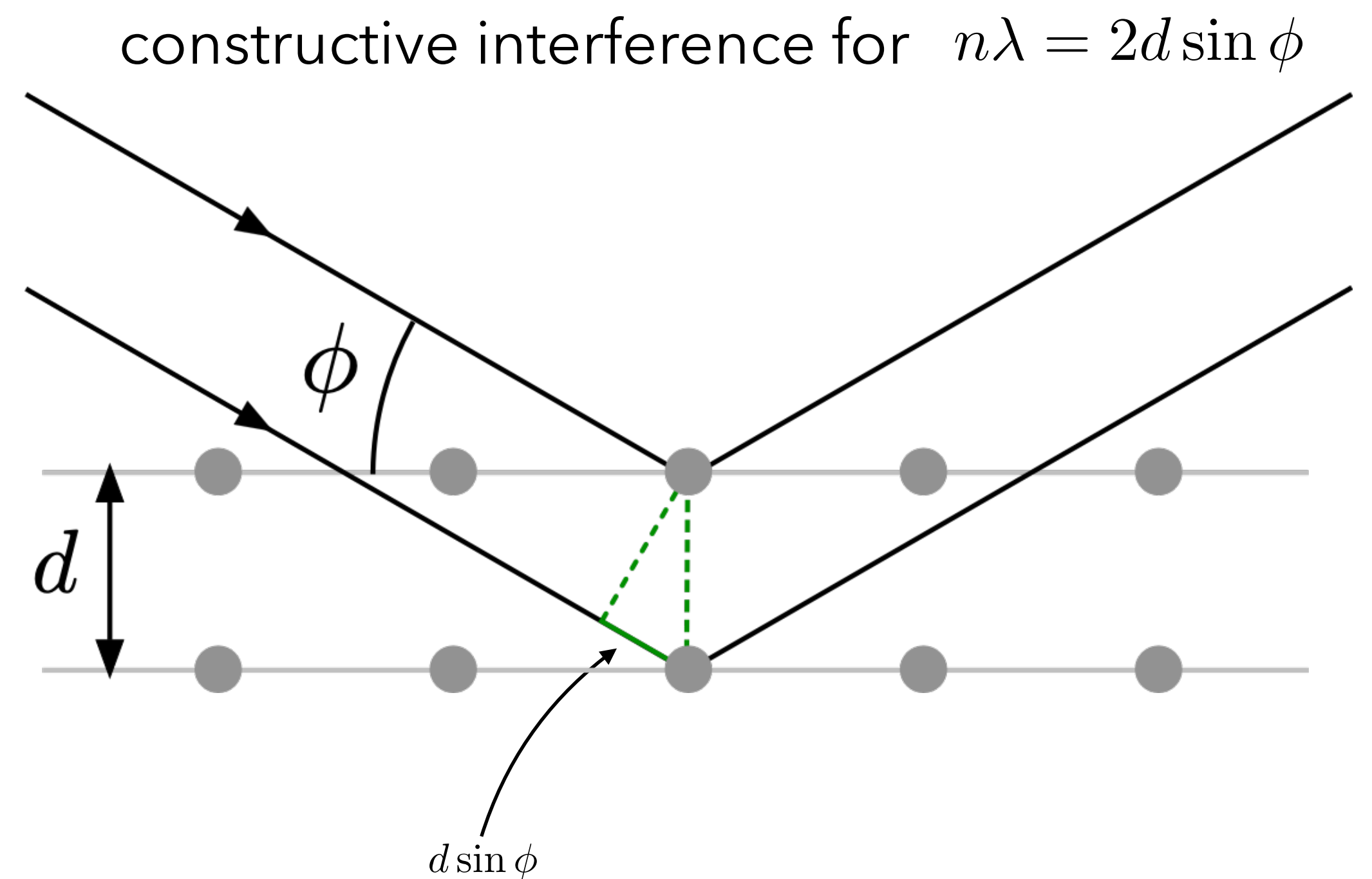
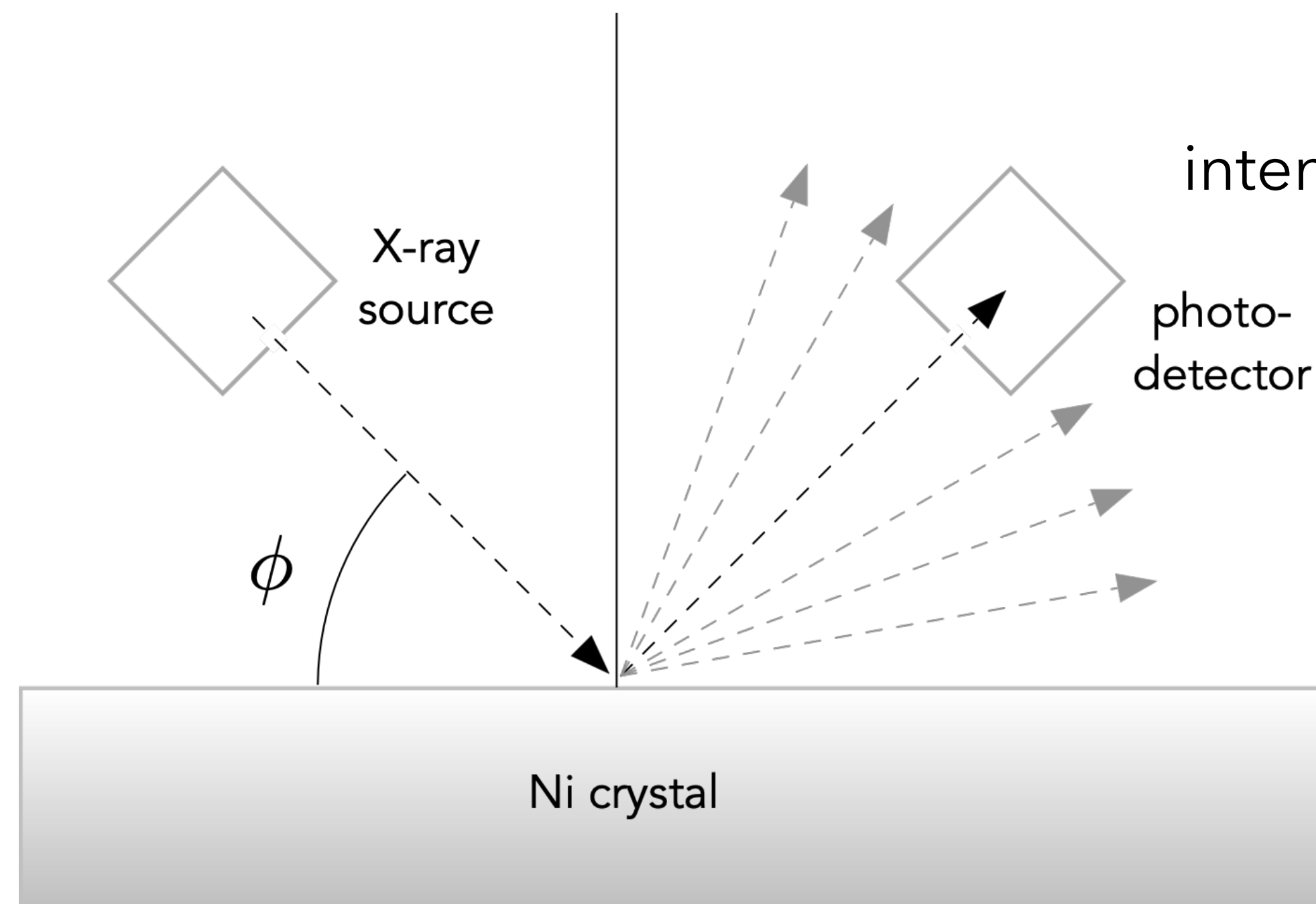
$$p = \sqrt{2E_{kin}m} = 1.031 \times 10^{-19} \text{ kg m/s}$$

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}}{1.031 \times 10^{-19} \text{ s kg m}} = 6.42 \times 10^{-15} \text{ m}$$

Wave character of matter

Experimental confirmation: **Davisson-Germer (1927)**

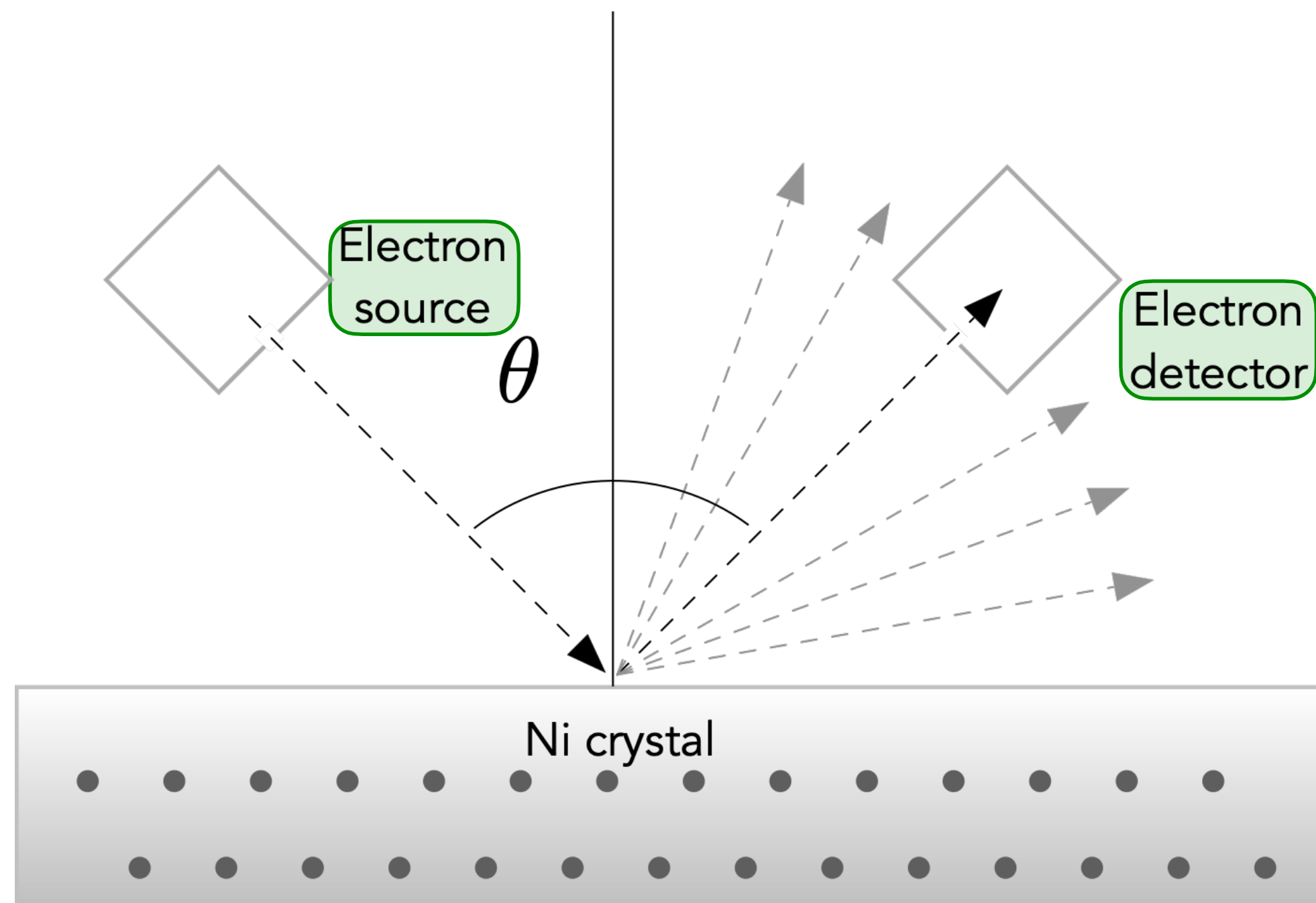
X-ray diffraction for structure determination



Wave character of matter

Experimental confirmation: **Davisson-Germer (1927)**

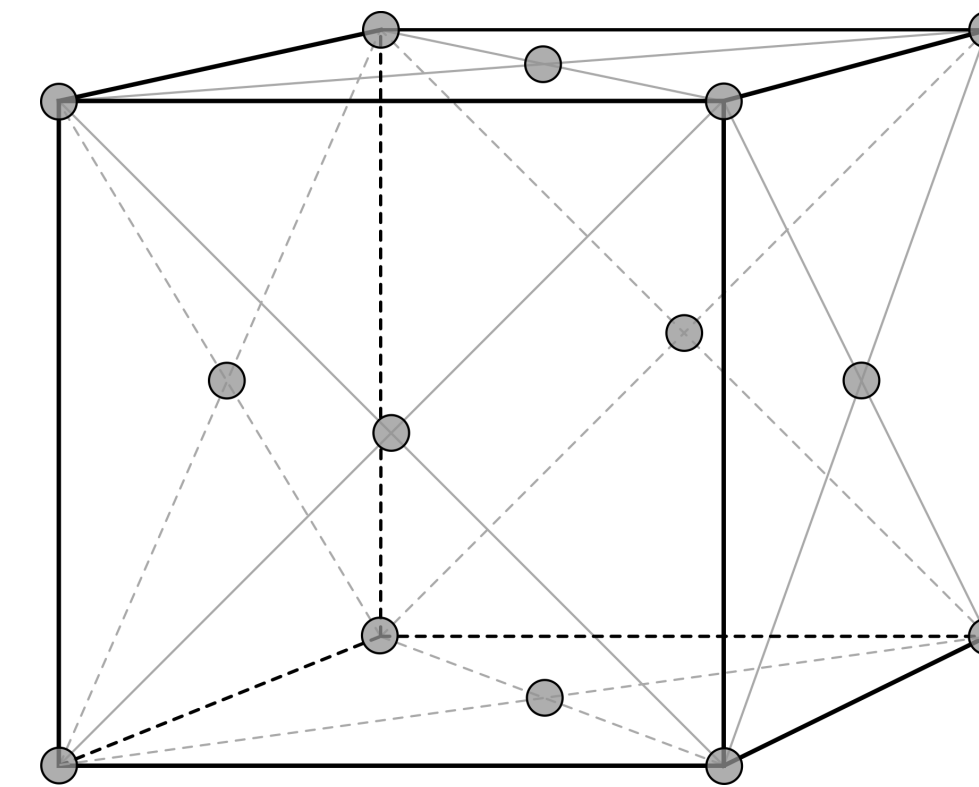
➔ Now with electrons



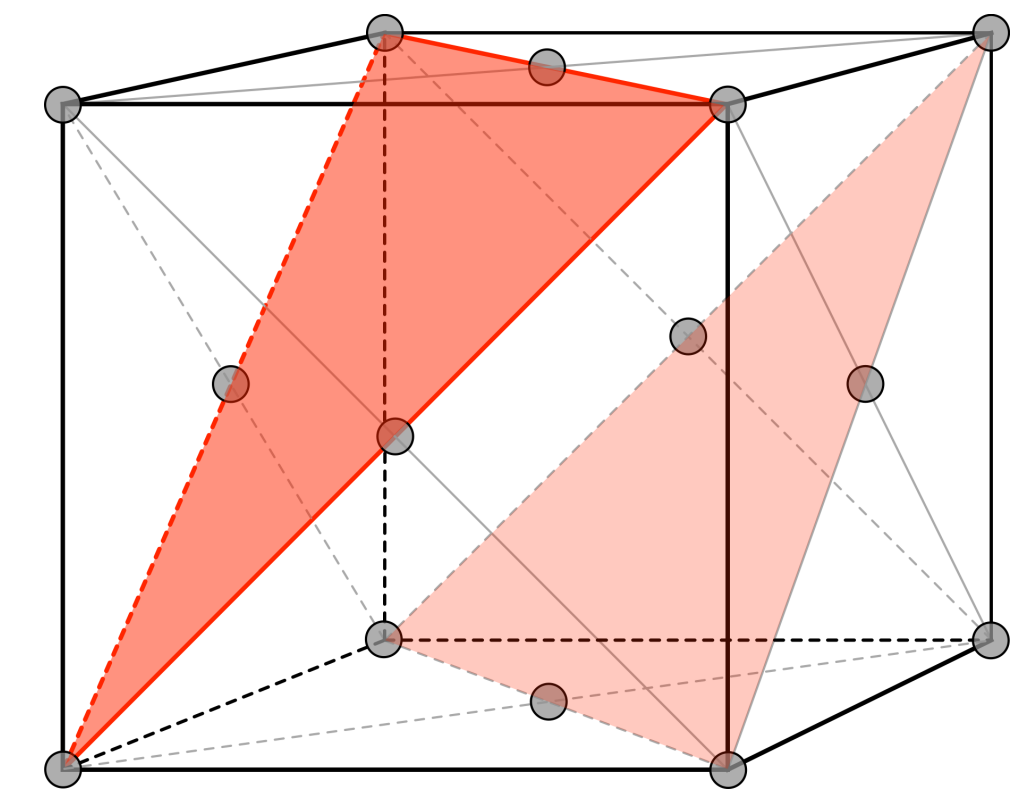
$$d = 0.091 \text{ nm}$$

$$\lambda = \frac{2d}{n} \sin \phi = \frac{2 \times 0.091 \text{ nm}}{1} \sin 65^\circ = 0.165 \text{ nm}$$

- Electrons with $E_{kin} = 54 \text{ eV}$
- maximal intensity at 50°
- minimal intensity at 35°



face-centered cubic

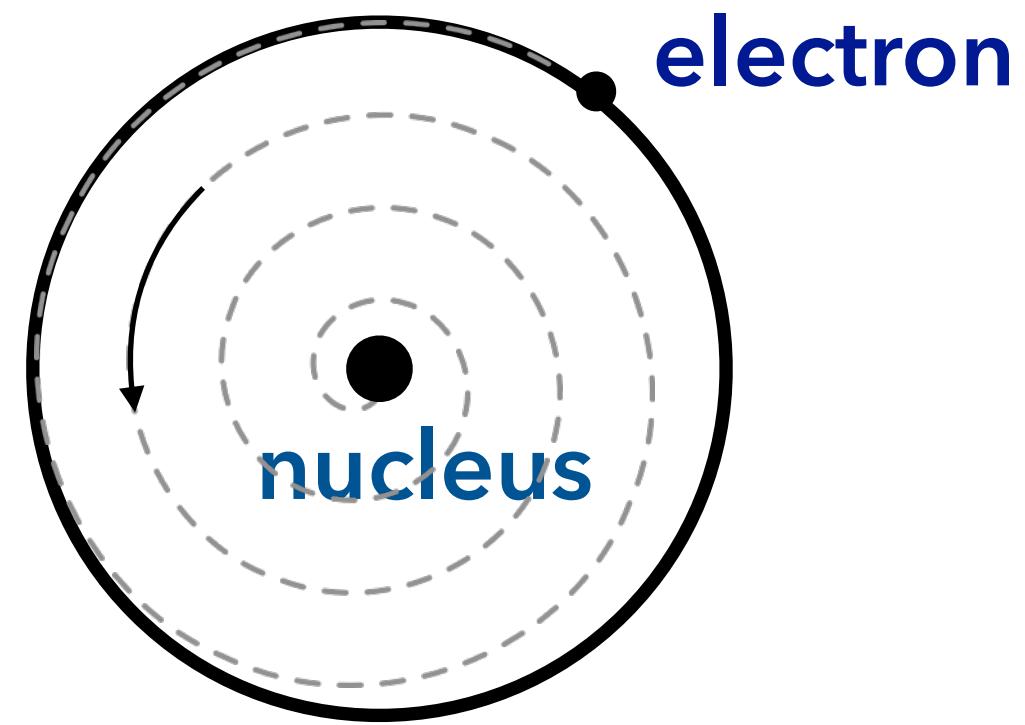


111-planes

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e E_{kin}}} = 0.167 \text{ nm}$$

Bohr's atom model

Rutherford



Accelerated charge emits energy (Maxwell)

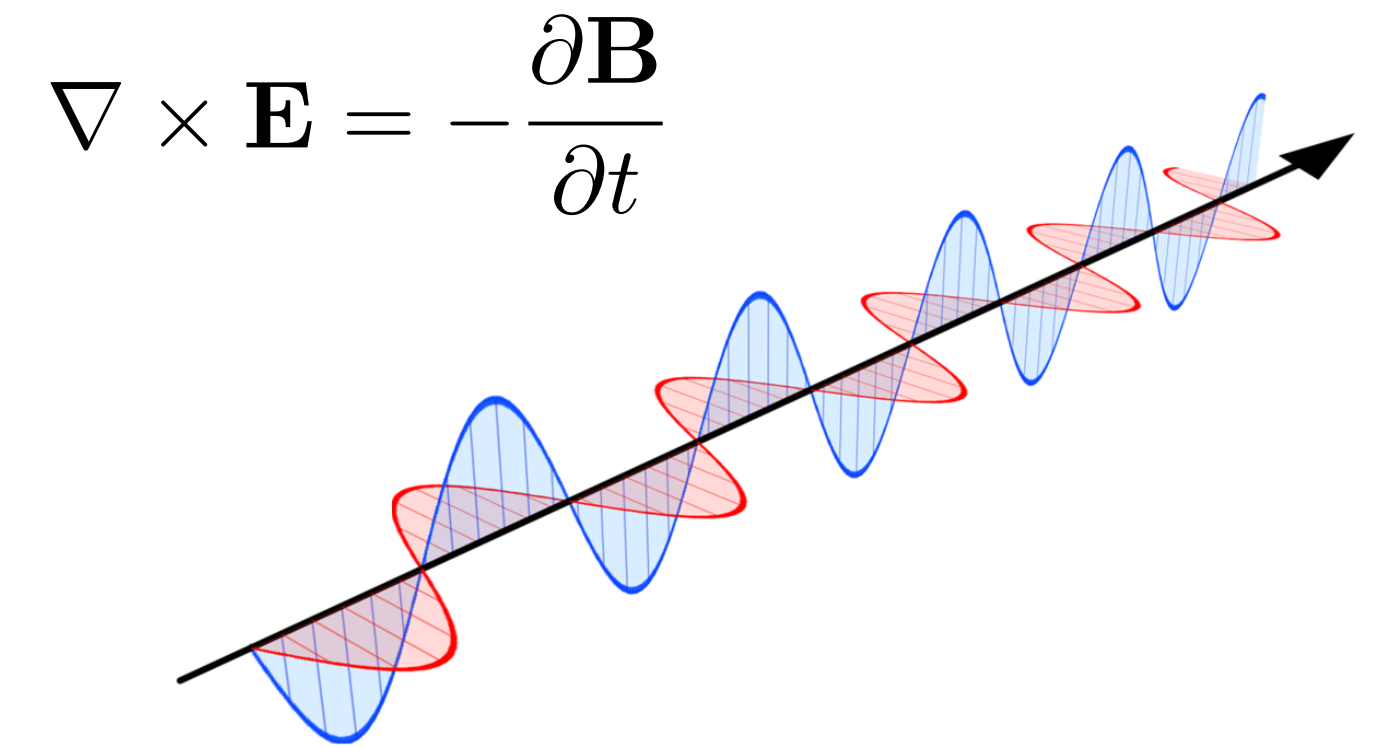


- Electron collapses onto the nucleus
- under the emission of radiation of continuously increasing frequency

but

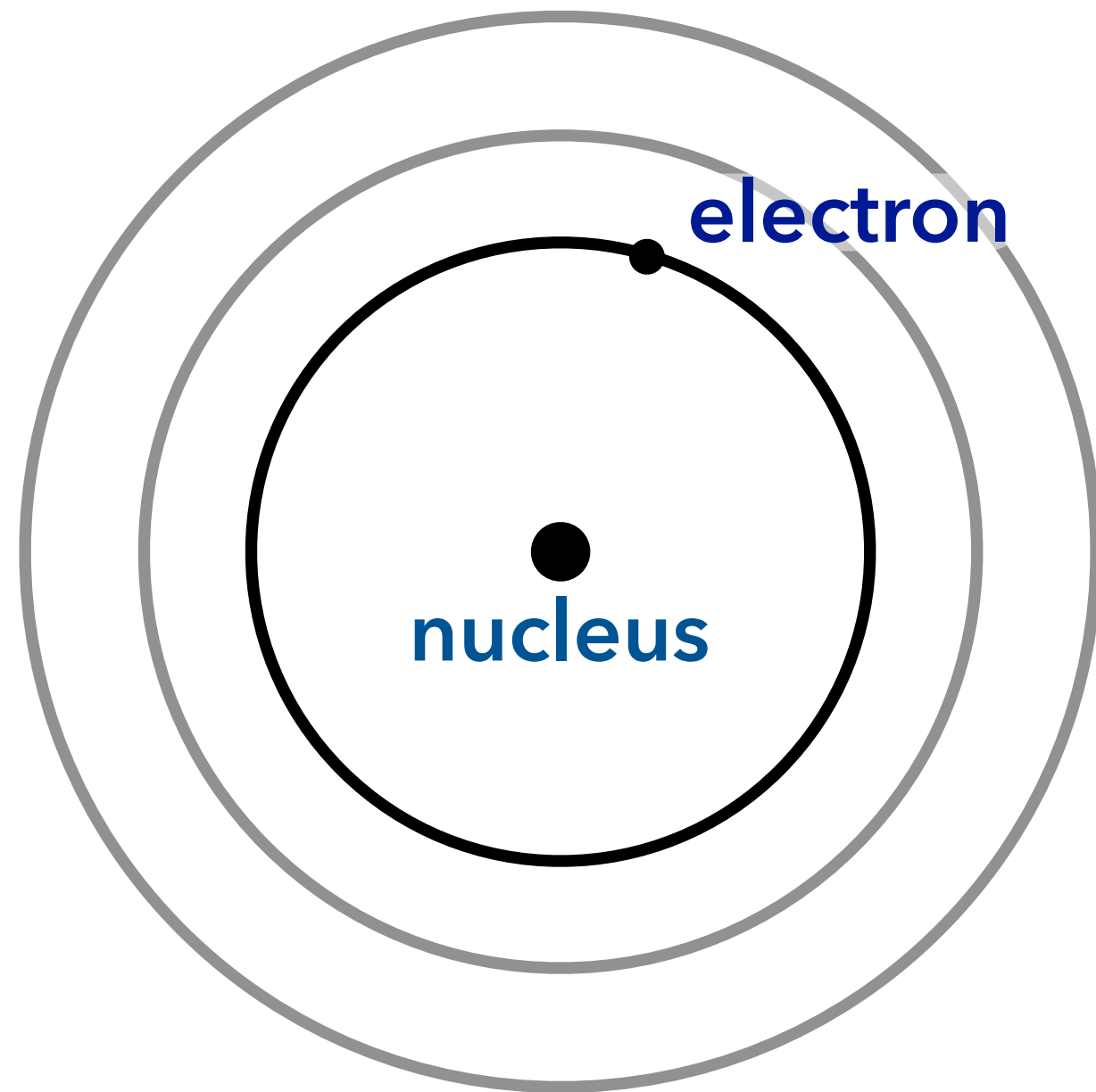
Atoms are stable

Discrete emission spectrum



Bohr's atom model

Bohr (1913):



1. Discrete orbits with energies:

$$E_1, E_2, E_3, \dots$$

2. Quantized orbital moment:

$$L = \frac{nh}{2\pi} = n\hbar$$

3. No emission unless electron jumps to another orbit:

$$E_2 - E_1 = h\nu$$

Electron in a circular orbit: $F = m_e a = m_e v^2 / r$

only electrostatic forces: $\frac{e^2}{4\pi\epsilon_0 r^2} = m_e \frac{v^2}{r}$

angular momentum: $L = m_e v r = n\hbar \Rightarrow m_e \frac{v^2}{r} = \frac{n^2 \hbar^2}{m_e r^3}$

$$\} \Rightarrow \frac{e^2}{4\pi\epsilon_0 r^2} = \frac{n^2 \hbar^2}{m_e r^3}$$

discrete radius: $r_n = \left(\frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} \right) n^2 = n^2 a_0$

Bohr radius: $a_0 = 0.0529 \text{ nm}$

Bohr's atom model

$$L = m_e v r = n \hbar \quad r_n = \left(\frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} \right) n^2 = n^2 a_0$$

discrete velocity: $v_n = \frac{n \hbar}{m_e r_n} = \left(\frac{e^2}{4\pi\epsilon_0} \right) \frac{1}{n \hbar}$

$$E = \frac{1}{2} m_e v^2 - \frac{1}{4\pi\epsilon_0} \frac{e^2}{r_n}$$

from $\frac{e^2}{4\pi\epsilon_0 r_n^2} = m_e \frac{v^2}{r_n}$ follows $\frac{1}{2} m_e v^2 = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \frac{e^2}{r_n}$

discrete energy: $E_n = -\frac{e^2}{8\pi\epsilon_0} \frac{1}{r_n} = -\frac{\mathcal{R}}{n^2}$

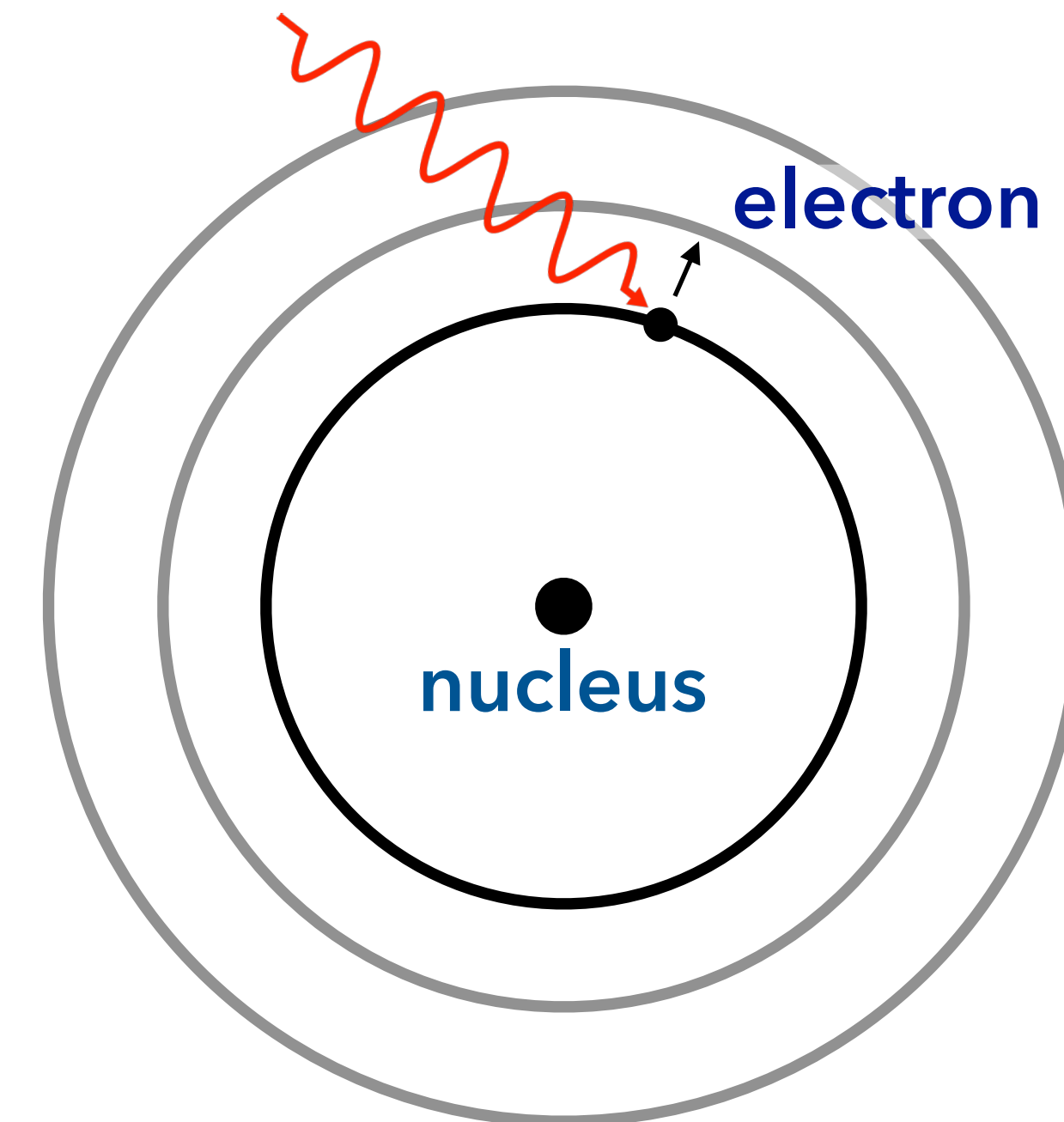
with $\mathcal{R} = \frac{m_e}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2$

$$\Delta E = -\mathcal{R} \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right) = 10.2 \text{ eV for the lowest transition}$$

fine structure constant:

$$\alpha = \frac{v_1}{c} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{\hbar c} = \frac{1}{137}$$

$$\Rightarrow E = -\frac{1}{2} \frac{e^2}{4\pi\epsilon_0 r_n}$$

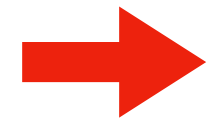


Rydberg constant:

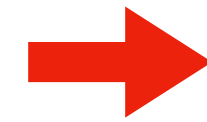
$$\mathcal{R} = 13.6 \text{ eV}$$

Bohr's atom model

de Broglie: $p = \frac{h\nu}{c} = \frac{h}{\lambda}$



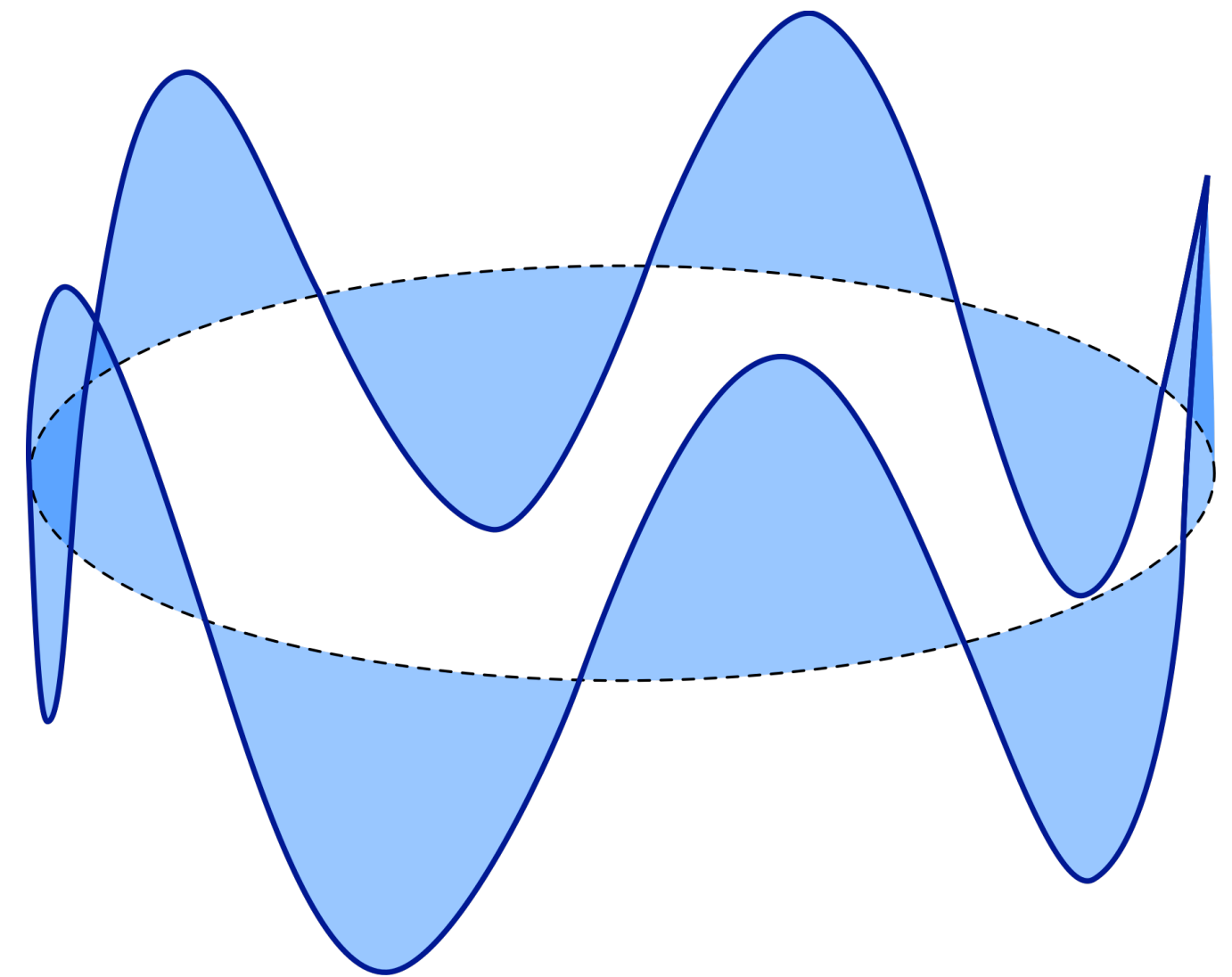
$$\frac{h}{\lambda} = \frac{nh}{2\pi r_n}$$



$$n\lambda = 2\pi r_n$$

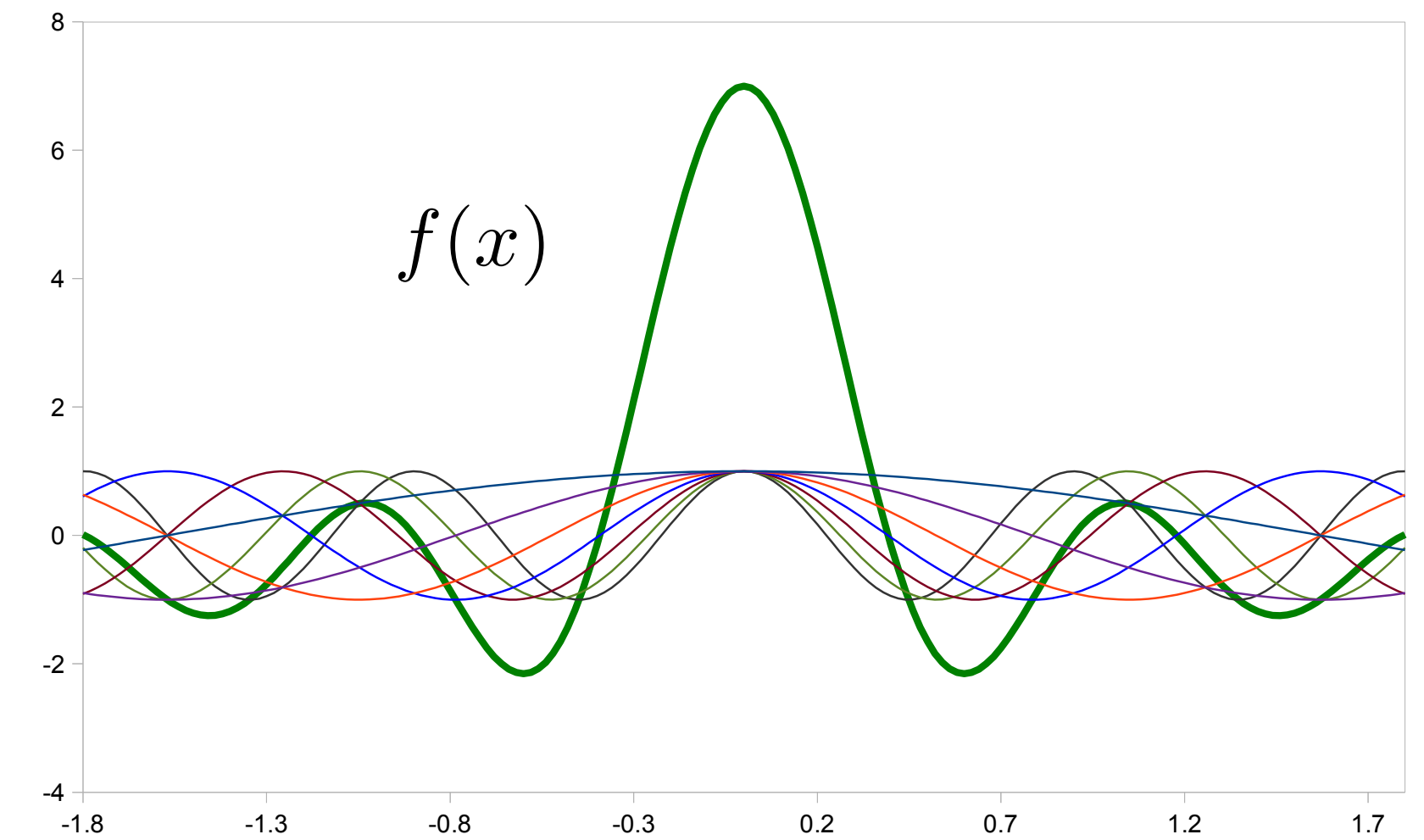
Bohr: $L = n\hbar$

Electron can be described by a standing wave on a circle



Wave packets

Fourier Series:
$$f(x) = \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$



$$a_{1\dots 7} = 1 \quad a_{n>7} = 0 \quad b_n = 0$$

One-dimensional plane wave: $\psi(x, t) = Ae^{i(kx - \omega t)}$

A : amplitude

k : wave vector

ω : angular frequency

Wave packet by superposition:
$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A(k) e^{i(kx - \omega t)} dk \quad (\text{Fourier transformation})$$

Wave packets

Wave packet by superposition: $\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A(k) e^{i(kx - \omega t)} dk$

at $t = 0$: $\psi_0(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A(k) e^{ikx} dk$

with $A(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \psi_0(x) e^{-ikx} dx$

$$x \rightarrow 0 \Rightarrow e^{ikx} \rightarrow 1$$

constructive interference

$$|x| \gg 0 \Rightarrow e^{ikx} \text{ can take any value}$$

destructive interference

Physical interpretation:

$\psi_0(x)$ wave function, amplitude

$|\psi_0(x)|^2$ probability density for finding the particle at position x

$P(x)dx = |\psi_0(x)|^2 dx$ probability for finding the particle between x and dx



Born interpretation of the wave function

Wave packets

but what about $A(k)$?

$$\int_{-\infty}^{+\infty} |\psi_0(x)|^2 dx = \int_{-\infty}^{+\infty} |A(k)|^2 dk = 1$$

$A(k)$ wave function, amplitude

$|A(k)|^2$ probability density for finding
the particle with wave vector k

$P(k)dk = |A(k)|^2 dk$ probability for finding the particles
wave vector between k and dk

Wave packets

Example of a localized wave packet:

$$\psi_0(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A(k) e^{ikx} dk$$

with $A(k) = \left(\frac{a^2}{2\pi}\right)^{1/4} e^{-\frac{1}{4}a^2(k-k_0)^2}$



$$\psi_0(x) = \left(\frac{2}{\pi a^2}\right)^{1/4} e^{-x^2/a^2} e^{ik_0 x}$$

$$\psi_0(x)^2 \propto e^{-2x^2/a^2} \Rightarrow 2\sigma^2 = \frac{a^2}{2} \Rightarrow \Delta x = \frac{1}{2}a$$

$$A_0(k)^2 \propto e^{-a^2(k-k_0)^2/2} \Rightarrow 2\sigma^2 = \frac{2}{a^2} \Rightarrow \Delta k = \frac{1}{a}$$

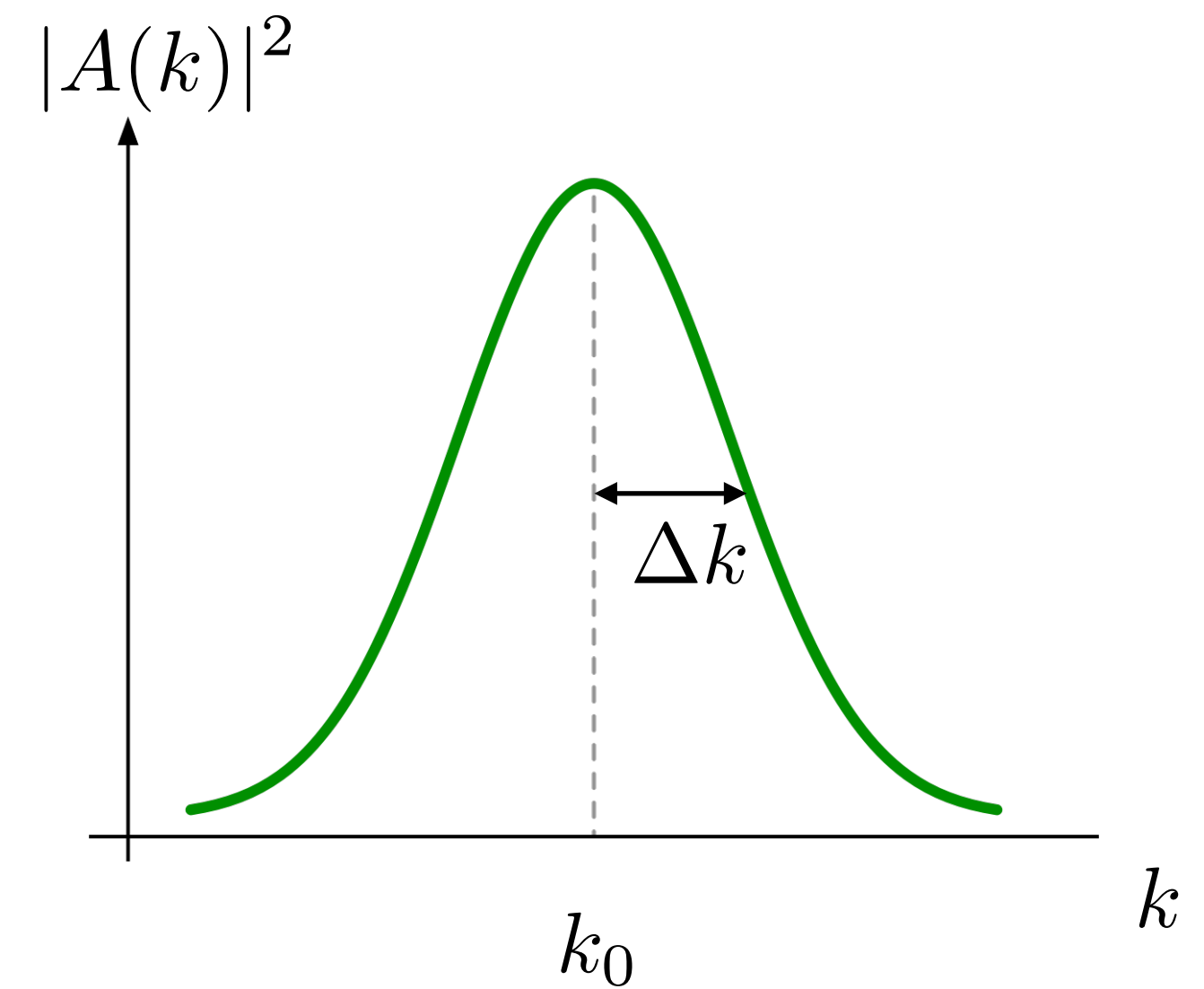
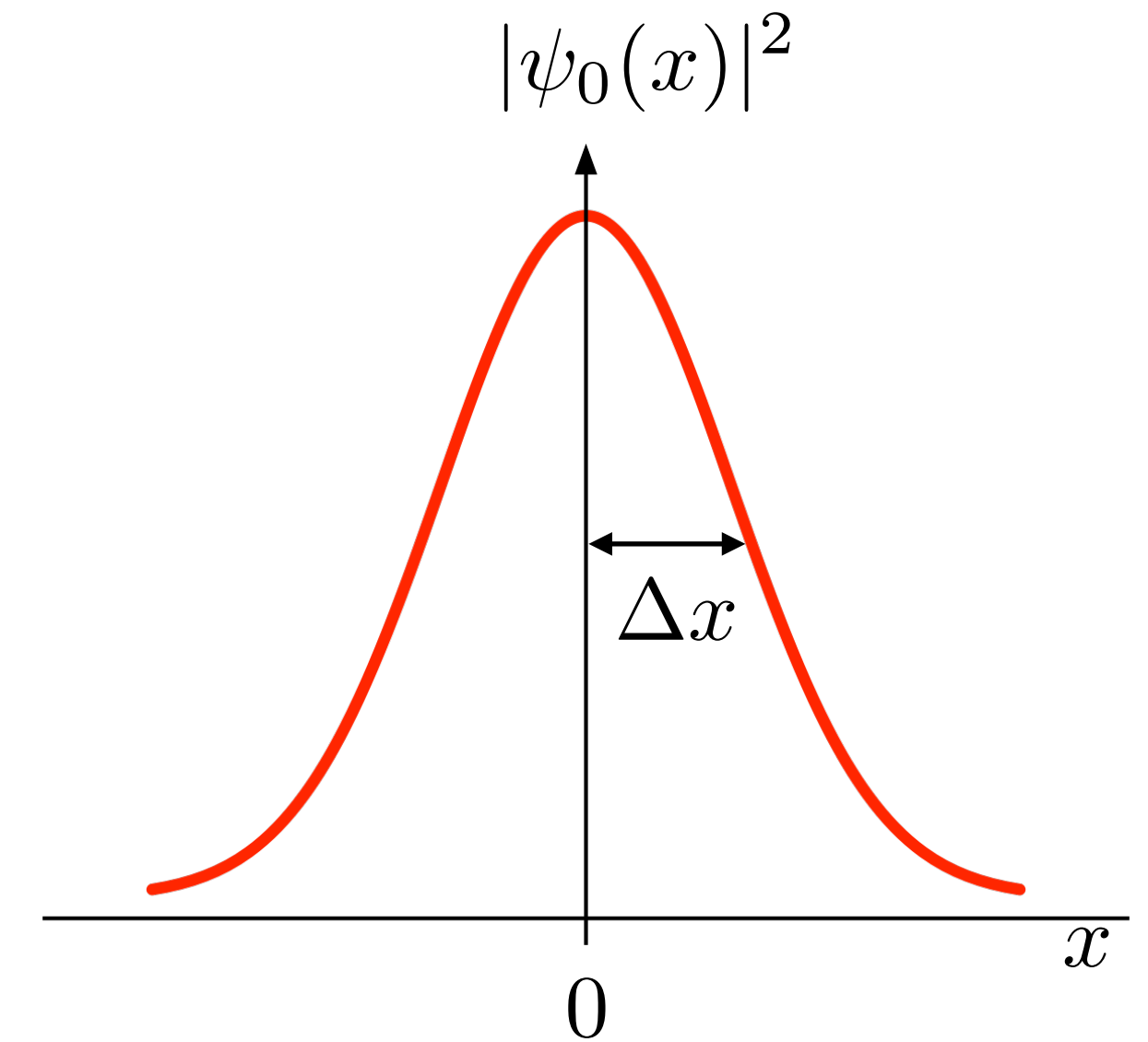
$$\Delta x \Delta k = \frac{1}{2}$$



$$\Delta x \Delta p = \frac{\hbar}{2}$$

$$k = \frac{p}{\hbar}$$

Two Gaussian wave packets



Time evolution

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A(k) e^{i(kx - \omega t)} dk$$

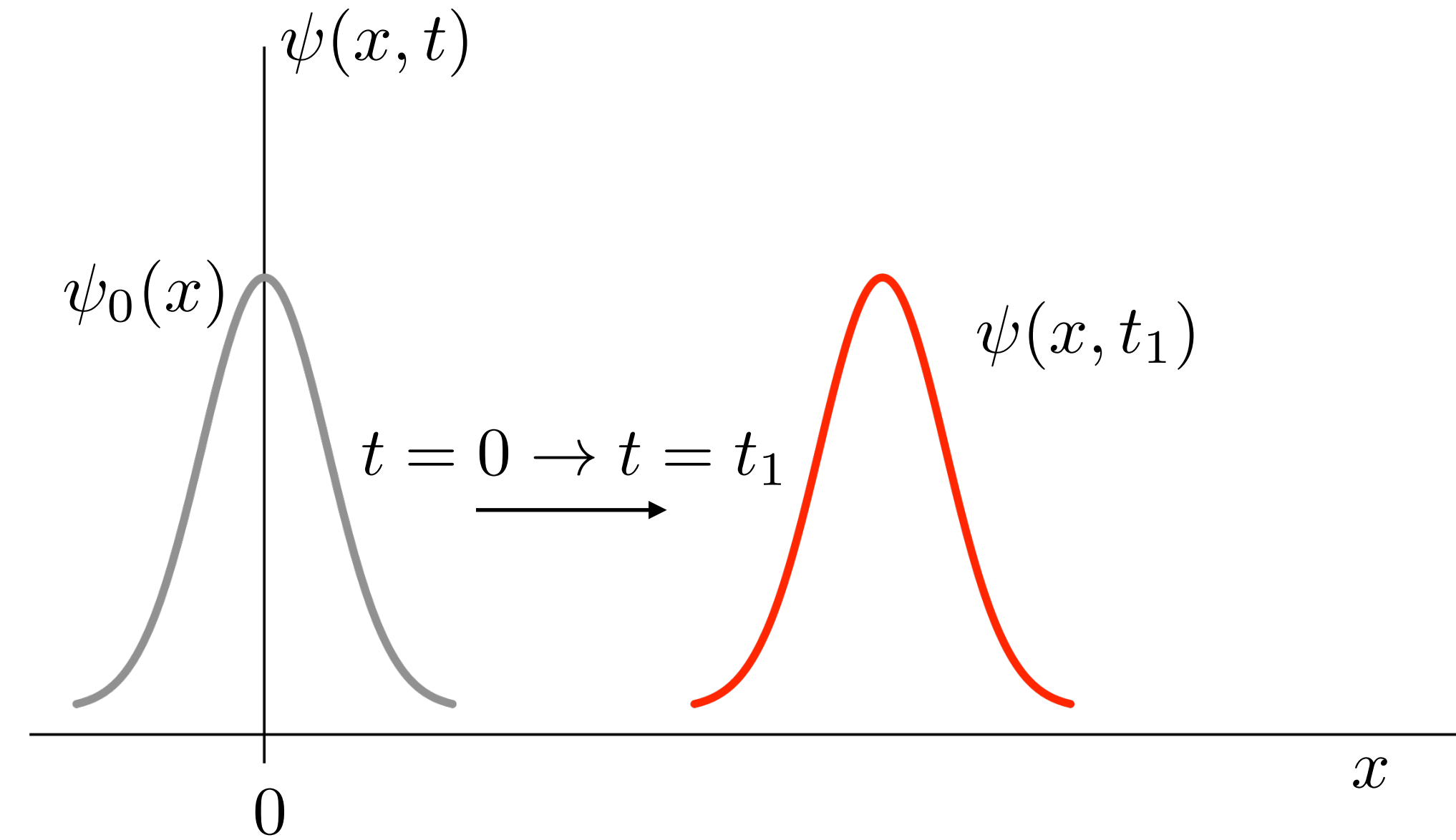
propagation in a non-dispersive medium: $\omega = v_0 k$

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A(k) e^{ik(x - v_0 t)} dk$$



$$\longrightarrow \psi(x, t) = \psi_0(x - v_0 t)$$

$$\psi_0(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A(k) e^{ikx} dk$$



Time evolution

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A(k) e^{i(kx - \omega t)} dk$$

propagation in a dispersive medium: $\omega = \omega(k) = \omega(k_0) + (k - k_0) \left. \frac{d\omega(k)}{dk} \right|_{k=k_0} + \frac{1}{2} (k - k_0)^2 \left. \frac{d^2\omega(k)}{dk^2} \right|_{k=k_0} + \dots$

linear approximation:

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} e^{i[k_0 x - \omega(k_0) t]} \int_{-\infty}^{+\infty} A'(k - k_0) e^{i(k - k_0)[x - \frac{d\omega(k)}{dk} t]} dk$$

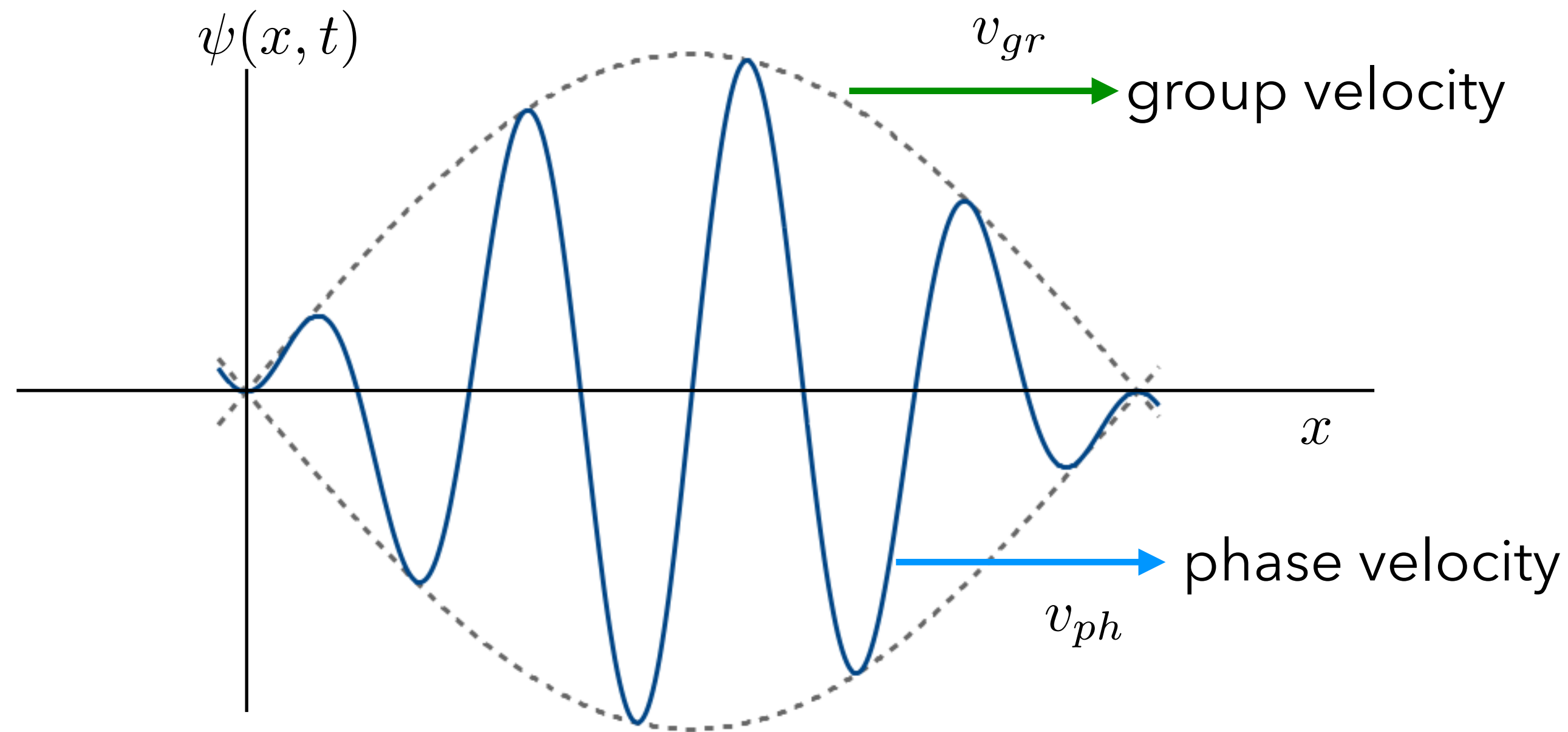
$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} e^{ik_0(x - v_{ph} t)} \int_{-\infty}^{+\infty} A'(k - k_0) e^{i(k - k_0)(x - v_{gr} t)} dk$$

$$v_{gr} = \frac{d\omega(k)}{dk} \quad \text{group velocity: moving wave packet}$$

$$\begin{array}{l} E = \hbar\omega \\ p = \hbar k \end{array} \quad \rightarrow \quad v_{gr} = \frac{dE}{dp} = \frac{d}{dp} \frac{p^2}{2m} = \frac{p}{m} \quad \rightarrow \quad \text{classical velocity of a particle}$$

$$v_{ph} = \omega(k_0)/k_0 \quad \text{phase velocity: individual waves in the packet}$$

Time evolution



$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} e^{ik_0(x-v_{ph}t)} \int_{-\infty}^{+\infty} A'(k-k_0) e^{i(k-k_0)(x-v_{gr}t)} dk$$

$$= e^{ik_0(x-v_{ph}t)} \psi_0(x-v_{gr}t) e^{-ik_0(x-v_{gr}t)} \quad \text{with} \quad \psi_0 = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A(k) e^{ik(x-v_{gr}t)} dk$$

No distortion, only phase modulation

Time evolution

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A(k) e^{i(kx - \omega t)} dk$$

linear approximation:

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} e^{ik_0(x - v_{ph}t)} \int_{-\infty}^{+\infty} A'(q) e^{iq(x - v_{gr}t)} dq$$

quadratic approximation: $\psi(x, t) = e^{ik_0(x - v_{ph}t)} f(x, t)$

with $f(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A'(q) e^{iq(x - v_{gr}t)} e^{-iq^2 \alpha t} dq$

$$q = k - k_0 \quad \alpha = \frac{d^2 \omega(k)}{dk^2}$$

Gaussian wave packet (*revisited*): $A(k) = \left(\frac{a^2}{2\pi} \right)^{1/4} e^{-\frac{1}{4} a^2 (k - k_0)^2}$

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \left(\frac{a^2}{2\pi} \right)^{1/4} e^{ik_0(x - v_{ph}t)} \int_{-\infty}^{+\infty} \exp \left[iq(x - v_{gr}t) - \left(\frac{a^2}{4} + i\alpha t \right) q^2 \right] dq$$

Time evolution

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \left(\frac{a^2}{2\pi} \right)^2 e^{ik_0(x-v_{ph}t)} \int_{-\infty}^{+\infty} \exp \left[iq(x - v_{gr}t) - \left(\frac{a^2}{4} + i\alpha t \right) q^2 \right] dq$$

$$|\psi(x, t)|^2 = \frac{1}{\sqrt{2\pi}\Delta x(t)} \exp \left[-\frac{(x - v_{gr}t)^2}{2(\Delta x(t))^2} \right] \quad \text{with} \quad \Delta x(t) = \frac{a}{2} \sqrt{1 + \frac{16\alpha^2}{a^4} t^2}$$

for a free particle: $\alpha = \frac{1}{2} \frac{d^2\omega(k)}{dk^2} = \frac{1}{2} \frac{d^2}{dk^2} \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m} \Rightarrow \Delta x(t) = \frac{a}{2} \sqrt{1 + \frac{4\hbar^2 t^2}{m^2 a^4}}$

